

AI-Powered Collections Strategy

Leveraging Agentic AI for Scalable, Fair, and Effective Debt Management at Geldium

Executive briefing for the Head of Collections

How the System Works: Four Steps

STEP 1

Inputs

- Repayment history
- Credit utilization, debt-to-income
- Credit score, income, tenure

Age and Location are excluded

STEP 2

Decision

- Risk score (logistic regression)
- Business rules as safety limits
- Low, Medium or High band

Every score has reasons

STEP 3

Action

- Low: automated reminder
- Medium: early support
- High: agent call, hardship review

Support first, never penalties

STEP 4

Learning

- Compare with a holdout group
- Monthly drift and fairness checks
- Human sign-off to retrain

Closes the loop

Risk Bands Decide the Action

Risk band	Example trigger	Action	Who decides
Low	Score below the Medium cut-off	Automated payment reminder	System
Medium	Higher score, or utilization of 30% or more	Supportive SMS or email with repayment options	System sends; agent reviews replies
High	Score above the High cut-off, or misses continue after outreach	Agent call and hardship plan review	Human agent

Go-live gate: the model must reach AUC of 0.70 or more and pass fairness checks. On current data it scores near chance (about 0.45), so the first release is a pilot.

Role of Agentic AI

Autonomous	Human Oversight
Refresh and validate customer data	Approve hardship and repayment plans
Score customers and assign risk bands	Handle high-risk and complex cases
Send standard reminders to low-risk customers	Any fee, credit-limit or account change
Pick channel and timing within approved limits	Resolve disputes and complaints
Monitor drift and raise fairness alerts	Approve model and threshold changes

Rule of thumb: automate routine, reversible, low-impact steps; keep people on anything that changes a customer's terms.

Responsible AI Guardrails

Fairness

Age and Location excluded from decisions. Quarterly checks of flag and error rates by group (ratio of 0.80 or better).

Explainability

Every score comes with plain-language reasons, such as "flagged: 45% of limit used".

Compliance

Built to meet ECOA, GDPR and local contact rules, with legal review before launch.

Human control and audit

No automatic denials or penalties. Full decision log, a named owner and a kill switch.

Expected Business Impact

2 pts 90-day delinquency cut (17% to 15%)

0.70+ AUC needed to go live

0.80+ minimum fairness flag-rate ratio

Business KPIs

- Delinquency rate down, cure rate up
- Cost per dollar collected down
- More accounts handled automatically
- Model and fairness gates met

Customer outcomes

- Earlier, supportive contact
- Fewer harsh or repeated calls
- Same fair treatment for similar customers
- Clear reasons for each decision

Targets to be confirmed in the pilot; they are goals, not forecasts.

Next Steps: Pilot Before Scaling

Month 1

Validate

Confirm data definitions, set the sample size, and complete legal and fairness review.

Months 2 to 4

Pilot

Contact half of eligible customers; keep half as a comparison group.

Months 5 to 6

Decide

Check AUC, fairness and delinquency results. Scale only if all pass.

Decision requested: approve the 6-month pilot, with go-live only if the model and fairness gates are met.